

Generative Adversarial Networks (GANs) for Image Generation

Objective: To understand and implement a Generative Adversarial Network (GAN) to generate realistic images. Train a GAN on a selected dataset and analyze its performance based on generated samples and loss curves.

Problem Statement: Given a dataset of images, we aim to develop a GAN that can generate synthetic images that closely resemble real ones. The generator should learn to create high-quality images, while the discriminator should be trained to distinguish real images from generated ones.

- Download the **MNIST** dataset, which consists of 28x28 grayscale images of handwritten digits (0-9)
- Normalize images to $[-1, 1]$ and also visualize a few samples of the data.
- Define the Generator and Discriminator models.
- Train the Discriminator: Use real images and generated images, compute loss, and update weights.
- Train the Generator: Generate images and try to fool the Discriminator.
- Evaluate the trained GAN by generating new images.
- Plot the Loss Curves.
- Analyze the result.
- If the accuracy of the model is not good, give some probable solutions and show the comparative result.

Deliverables:

- Python code for all the tasks mentioned above. Plots showing accuracy trends.
- A report summarizing your analysis, observations, and conclusions.